

Be ready to manage stress “Before” and “After” a critical event. What the EEG and autonomic correlates tell us

Laura Angioletti^{a,b,*}, Katia Rovelli^{a,b}, Michela Balconi^{a,b}

^a International research center for Cognitive Applied Neuroscience (IrcCAN), Università Cattolica del Sacro Cuore, Milan, Italy

^b Research Unit in Affective and Social Neuroscience, Department of Psychology, Università Cattolica del Sacro Cuore, Milan, Italy

ARTICLE INFO

Keywords:

Stress management
RTs
EEG
Autonomic measures
Emotion regulation
Interview

ABSTRACT

This study examined behavioral, electrophysiological (EEG), and autonomic responses to stress during the preparation and speech stages of five discourses among 26 adults. Participants underwent an increasingly stressful job-interview based on a modified Trier Social Stress Test, receiving feedback from an evaluative board. Findings showed increased RTs, higher cardiovascular responses [Pulse Volume Amplitude (PVA), and Heart Rate Variability (HRV)] and generalized increases in EEG frequency bands (delta, theta, alpha, beta, gamma) during the speech compared to the preparation stage. The rising emotional salience of the discourses induced a negativity bias and extensive low-frequency band activation (delta and theta) across the scalp in response to emotional demands. High-frequency bands exhibited a plateau effect, indicating less cognitive involvement as the discourses progressed. In our opinion, a possible interpretation is that this effect could be due to habituation mechanisms or coping strategies. Autonomic results revealed significant variations in PVA, with higher levels during the first discourse preparation, indicating substantial cognitive effort. Despite increased emotional arousal, participants managed stress effectively, as evidenced by increased HRV during the speech stage. Overall, during progressively increasing ecological psychosocial stress, individuals displayed marked emotional reactions in terms of low-frequency bands and cardiovascular indices, particularly during the first speeches rather than the preparation stages of an interview.

1. Introduction

How do individuals react during progressively increasing acute psychosocial stress (such as during an interview in front of an evaluation panel)?

Psychosocial stressors encompass a particular category of stress-inducing stimuli characterized as challenging or demanding elements originating from social interactions, primarily due to their unfamiliar, unpredictable, or uncontrollable nature, or the existence of social evaluation pressures. These stressors are strictly related to social interactions and arise from the cognitive appraisal or understanding of a psychosocial stressor, which strains or surpasses an individual's coping capacities (Vanhollebeke et al., 2022).

Several studies examining reactions to psychosocial stressors do not analyze the entire stress response, instead concentrating on phases of it. When categorizing the various phases of an acute stress response based on their timing in relation to the current stressor, three distinct phases emerge: the *anticipatory phase*, the *reactive phase*, and the *recovery phase*

(Vanhollebeke et al., 2022). Firstly, the *anticipatory phase* coincides with the moment in which an individual becomes aware of the future stressor but has not yet encountered it directly (Engert et al., 2013). Subsequently, the *reactive phase*, refers to the period during which an individual faces directly the stressor. Finally, the third phase is the *recovery phase*, which commences immediately following the cessation of exposure to the stressor.

In the psychological literature, one of the most well-known standardized stress-inducing protocol is the Trier Social Stress Test (TSST; (Kirschbaum, Pirke, & Hellhammer, 1993) which was designed to evoke a psychobiological stress response in humans. This test involves two components: a spontaneous speech and a mental arithmetic task, both performed in front of an audience. Substantial evidence proved the efficacy of the TSST in eliciting a multidimensional stress response (Campbell & Ehler, 2012; Dickerson & Kemeny, 2004). Literature identifies the TSST as a robust psychosocial stressor, promptly activating all major physiological stress pathways in the central nervous system (CNS) and autonomic nervous system (ANS), comprising its sympathetic

* Corresponding author at: Department of Psychology, Università Cattolica del Sacro Cuore, Largo Gemelli 1, Milan 20123, Italy.

E-mail address: laura.angioletti1@unicatt.it (L. Angioletti).

<https://doi.org/10.1016/j.bandc.2024.106244>

Received 16 October 2024; Received in revised form 18 November 2024; Accepted 3 December 2024

Available online 9 December 2024

0278-2626/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

and parasympathetic branches.

By considering this specific social stressor and building on the knowledge derived from the previously described neuroscientific studies on the TSST (Barzegar, Jahromi, Meftahi, & Hafez, 2023; Boesch et al., 2014), it becomes interesting to distinguish and analyze two distinct inner stages of this acute stress reactive phase, that are the *preparation stage* – where the individual has been exposed to the stress and is getting ready to perform – and the actual *speech stage* of the interview – corresponding to the performance itself. Moreover, the dynamics of an interview as an ecological and realistic condition of daily life, might foresee the exposure to increasing emotionally charged demands and/or positive/negative feedback, which may imply a different emotional regulation and, to our knowledge, have not been taken into consideration in previous studies.

Employing a comprehensive neuroscientific approach that integrates multiple methods, such as electrophysiological (EEG) frequency bands and autonomic data (autonomic nervous system, ANS), enables the exploration of diverse body-and-mind activation patterns in individuals during the exposure to a social stressor, such as during an interview in front of an evaluation panel.

Delving into the functional significance and scalp localization of electrophysiological (EEG) frequency bands can offer valuable insights into how individuals respond to social evaluative stressors (Vanhollebeke et al., 2022). In a recent review by Vanhollebeke and colleagues (Vanhollebeke et al., 2022), eight studies examined the EEG correlates of various phases of the psychosocial stress response during the TSST, with a primary focus on the alpha and beta bands. Of these, two studies specifically reported the increase of beta band (Izhar et al., 2019) and frontal alpha asymmetry (Wang et al., 2015) during the reactive phase, i.e., during the task. Interestingly, a recent study by Barzegar and colleagues (Barzegar et al., 2023) reported a significant increase in delta band frequency during the TSST, while beta bands experienced a significant decrease, alongside minor reductions in theta and gamma oscillations, particularly in the frontal region. Consequently, this work showed that stress during the TSST led to heightened low-frequency activity and diminished high-frequency EEG oscillations.

Indeed, prior research considering such crucial and central phase – that is the reactive phase – of other psychosocial stressors showed an increase in EEG low-frequency bands (delta and theta band) in response to social feedback (Yao et al., 2020), and emotional processing during the reactive phase (Poppelaars, Klackl, Pletzer, & Jonas, 2021). A significant decrease of alpha band was observed during the reactive phase compared to a control condition (Al-Shargie et al., 2017b; Al-Shargie et al., 2016; Al-Shargie et al., 2017a; Al-Shargie et al., 2018; Hafeez, Shakil, Jangsher, & Effects, 2018; Minguillon, Lopez-Gordo, & Pelayo, 2016; Yao et al., 2020); on the contrary, Betti and colleagues (Betti et al., 2018) found an increase of alpha power in the reactive phase compared to the baseline. Differently from Barzegar and colleagues' study (Barzegar et al., 2023), other former works reported the significantly higher beta power in frontal electrodes during the reactive phase (Hafeez et al., 2018; Minguillon et al., 2016; Perrin et al., 2019). Thus, it might be valuable to consider both low- and high-frequency bands during the reactive phase of a psychosocial stressor.

Likewise, examining the activity of the ANS can offer valuable perspectives on how an individual's ANS responds to social stressors, even in term of stress management abilities. Biomarkers such as cortisol (Hellhammer & Schubert, 2012; Schoofs & Wolf, 2011) and heart rate (Hellhammer & Schubert, 2012; Nater et al., 2005) exhibit an increase parallel to the increase in stress levels during the TSST, while heart rate variability (HRV) decreases in reaction to the TSST (Nater et al., 2006; Strahler, Mueller, Rosenlocher, Kirschbaum, & Rohleder, 2010). So far, a decrease in HRV during TSST has only been observed during single settings (Nater et al., 2006; Strahler et al., 2010), while in Boesch and colleagues (Boesch et al., 2014) study a significant increase in the HRV response was found during a second exposure, suggesting that this kind of stress task seems to provoke a stronger parasympathetic inhibition

when readministered. Also, other cardiovascular indices (such as the Pulse Volume Amplitude, PVA) and electrodermal activity (EDA) indices can be informative of the fight or flight stress response and have been correlated with latent stress situations (Stemmler, 2004).

Thus, the objective of this study is to compare the EEG and autonomic correlates observed during the *preparation* and *speech* stages of a psychosocial stressor across progressively intensifying emotional demands, within a cohort of healthy participants.

To do so, it was employed a modified version of the TSST (Allen et al., 2017), named Social Stress Test (SST), where participants were asked to prepare and subsequently present five distinct discourses on progressively higher emotionally charged topic, thus requiring increasing emotional demands. Between the preparation and the speech stage, participants watched five video-auditory emotional stimuli (i.e., five dynamic videos portraying an examining board commenting on their performance) gradually characterized by higher negative emotional connotations during each discourse. Furthermore, the emotional intensity was heightened by the facial expressions of the evaluative board, which became progressively more negative with each video.

To gain a thorough understanding of how highly stimulating content affects cognitive and emotional performance as well as to explore the implicit emotional processes (Balconi, Grippa, & Vanutelli, 2015), individuals neurophysiological signals were monitored via EEG and autonomic data co-registration. Additionally, at the behavioral level, the emotional impact and cognitive workload was gauged by measuring the reaction times (RTs) during both the preparation and speech stages. RTs were collected with the objective of analyzing the emotional regulation of subjects and their capacity to effectively regulate and resist stress in different mental processes related to performance (preparation *versus* speech). RTs not only reflect the emotional reactions, but also the interaction of brain processes involved in planning, language processing, and executive control. For this reason, RTs can be considered a reliable measure of cognitive workload involved in the evaluation of preparation and speech performance. In line with the literature, the RTs required to prepare a discourse performance reflect a series of high-level cognitive functions, such as memory access, selection of relevant concepts, language generation, and supervision of the entire speech sequence (Elsei, Filmer, Galvin, Kurath, Vossoughi, Thomander, Zavodnik, & Kindt, 2020; Rönnerberg et al., 2013). Similarly, the RTs required for processing questions during the speech stage can reflect efficiency in accessing stored information and be an index of fluency and coherence in language production (Huang & Shanks, 2021).

Based on these premises, on a behavioral level, we expect an increase in RTs for the speech compared to the preparation stage, as the speech stage appears more salient on an emotional level and may require greater cognitive effort and emotion regulation abilities. However, the exposure to highly emotionally salient stimuli may induce an attentional bias corresponding to increased emotional salience (i.e., negativity bias; Balconi & Pozzoli, 2009), which we presume could manifest as a reduced RTs in concomitance with emotional escalation associated with the discourses.

Additionally, for the EEG data, we hypothesized an increase of the EEG delta band related to emotional processing (Barzegar et al., 2023) during the inner and most stressful stage of the interview, that is the speech stage, compared to the preparation stage. Moreover, during the speech compared to the preparation stage, we also expect an increase of theta band given the exposure to the social feedback (Yao et al., 2020). We expect that a distribution of the presence of these low frequency band waves more widespread across the entire scalp in speeches characterized by requests and social feedback of greater emotional salience, such as those characterizing the last discourses.

Regarding high-frequency bands (alpha, beta, and gamma band), in line with previous studies on the stress reactive phase (Betti et al., 2018; Hafeez et al., 2018; Minguillon et al., 2016; Perrin et al., 2019), we expect an increase of these frequency bands in relation to the most

stressful stage of the interview, that is the speech stage compared to the preparation stage. Since high frequency EEG (alpha, beta, and gamma) oscillations seem to be more commonly associated with brain states requiring attention and alertness (Vanhollebeke et al., 2022), we expect that this activation could mirror the cognitive workload needed for facing increasingly stressful situations and, specifically, we suppose that this effect may vary considering the increasing emotional salience of the different speeches, however maintaining a greater increase for the speech towards the preparation stage within the different speeches.

Regarding the ANS, we expect that with the increasing emotional demand of the five speeches there may be an increase in emotional automatic processes related to the fight or flight response (such as cardiovascular and EDA indices) during the speech compared to the preparation stage. Furthermore, it is supposed that HRV increases significantly during the speech versus preparation stage, signaling the most stressful (Boesch et al., 2014) and resource-requiring stress management stage of the SST.

2. Methods

2.1. Sample

The sample included 26 healthy young adults ranging from 22 to 28 years old (10 males and 16 females, Mean (M) age = 23.04, Standard Deviation (SD) age = 1.46). Exclusion criteria included: no previous or current history of psychiatric or neurological disorders, as well as the exclusion of individuals undergoing psychoactive drug therapies that could potentially affect central nervous system functioning.

Moreover, participants were required to have no clinically significant levels of stress or depression, and they should not have experienced any major stressful life events within the past six months. The Perceived

Stress Scale (PSS; (Cohen, Kamarck, & Mermelstein, 1983) and the Beck Depression Inventory (BDI, (Sica & Ghisi, 2007) were administered to exclude clinically significant levels of stress and depression in the selected sample (PSS score: M = 18.8, SD = 6.12; BDI score: M = 7.25, SD = 5.56). Furthermore, all participants had normal or corrected-to-normal hearing and vision. Prior to participation, written informed consent was obtained from all individuals involved in the study, adhering to the ethical guidelines outlined in the Declaration of Helsinki (2013).

2.2. Experiment procedure

The experiment was conducted in a controlled, moderately lit room. Participants were informed in advance that they would take part in a simulated and computerized job interview lasting approximately 30 min, with the presence of a fictitious evaluating board which was video recorded. Participants were also informed that the procedure would involve the application of EEG and autonomic measures recording.

Each participant was asked to sit facing a workstation consisting of a monitor and mouse, approximately 80 cm away from the screen, with feet firmly on the ground. After signing the informed consent form, electrodes for EEG were applied to the scalp of the participant, and sensors for detecting autonomic indices were applied to specific hand reference points. Subsequently, 120 s resting state baseline activity was recorded with eyes open and eyes closed. After the baseline, they were required to perform the SST, while EEG and autonomic data were recorded continuously. The experimental procedure lasted approximately 1 h (Fig. 1).

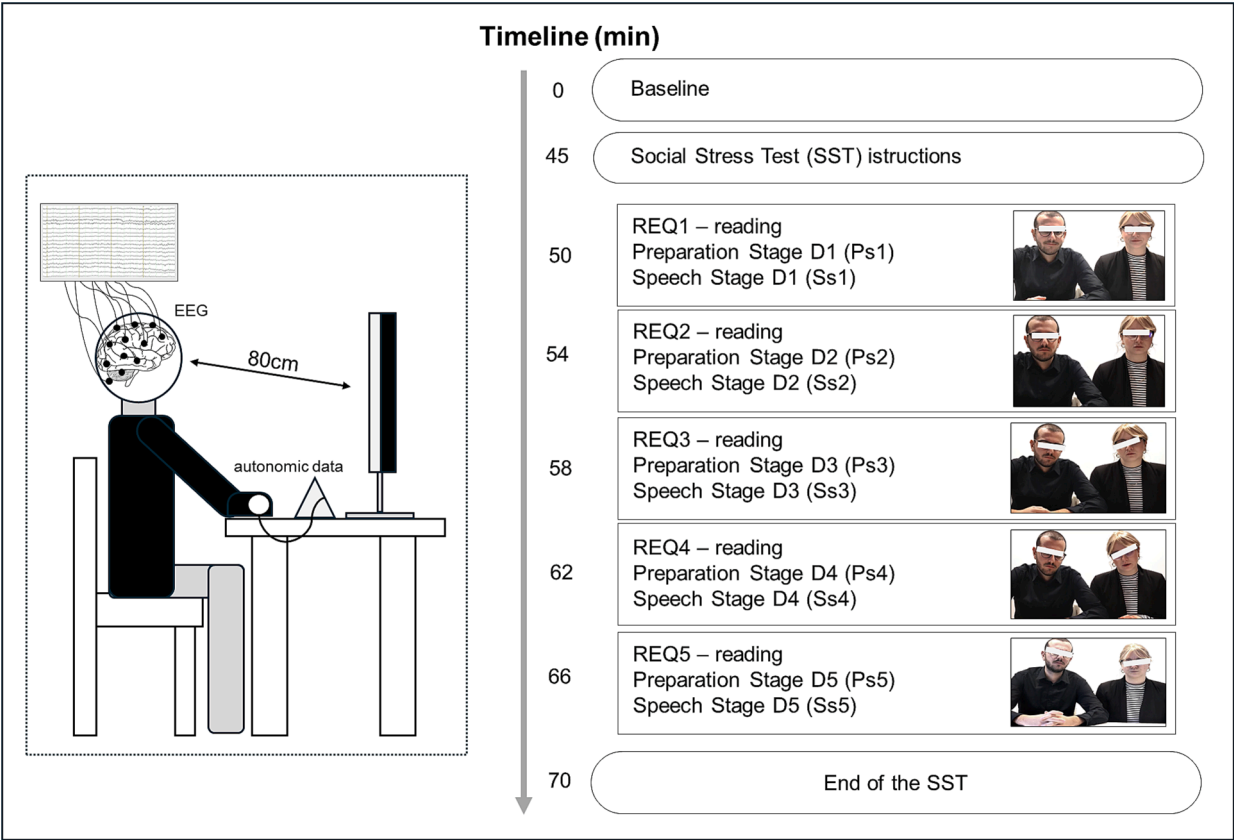


Fig. 1. Experimental procedure. The sequence of discourses within the experimental procedure is illustrated. EEG activity was monitored from the baseline throughout the task, as well as the recording of behavioral data.

2.3. Social stress test

The SST was administered through a web-based experiment management platform (Qualtrics XM platform; Qualtrics LLC, Provo, UT, USA). Before starting the experiment, each participant was required to read the following instructions:

“Imagine yourself in a competitive screening process for an internship or job role that holds significant importance to you. The organization’s representative reveals that only one position is vacant, and there are 23 other candidates vying for it. Your next step involves facing a selection interview, during which you’ll encounter a simulated panel of examiners. To ace this, you must ready yourself to prepare and present various discourses orally.”

Particularly, as specified in the assignment, participants were asked to prepare and present five different speeches (Discourse₁₋₅, D₁₋₅), each consisting of 4 steps: (i) reading the assignment (Assignments₁₋₅, A₁₋₅); (ii) speech preparation in 120 s (Preparation Stage₁₋₅, Ps₁₋₅); (iii) viewing the simulated video of the examining board (Video₁₋₅, V₁₋₅); (iv) delivering the speech in 60 s (Speech Stage₁₋₅, Ss₁₋₅).

Each Assignment was progressively manipulated from self-centered requests and involving individual decisions to increase the engagement and the stress levels (A₁: “prepare the best presentation of yourself”; A₂: “describe a situation where you encountered difficulty in making a decision”; A₃: “describe a situation where you made a difficult decision without any support”) to requests involving other-oriented decisions (A₄: “describe a situation where you made a decision taking full responsibility on behalf of others, without being able to consult them”; A₅: “describe a situation where you made a decision in disagreement with the rest of the group”).

Similarly, the emotional intensity, the level of engagement and stress of the video-auditory stimuli from the examining board was manipulated, with only the video of the first examining committee being positive (e.g., V₁: *friendliness*; V₂: *neutrality*; V₃: *boredom*; V₄: *increasing impatience*; V₅: *total aversion*). External judges rigorously examined and assessed the emotional stimuli presented in this study to ensure their consistency and reliability in eliciting the intended experimental outcomes, emphasizing coherence and the dependability of the desired results. Furthermore, the participants’ post-test reports revealed a correlation between the specificity of the videos and their escalating stress levels, underscoring the significant impact of the increasing emotional content on their psychophysiological responses. Additionally, the heightened intensity of negative emotions reproduces in the videos was associated with a greater induced stress.

2.4. Behavioural data acquisition and processing

Specifically, the RTs for the Ps and the Ss were collected using a time-monitoring function provided by the web-based experiment management platform employed (Qualtrics XM platform; Qualtrics LLC, Provo, UT, USA). A maximum time of 120 s was established for each preparation phase (Ps_{RTs} max), while a maximum time of 60 s was established for the speech stage (Ss_{RTs} max). In order to compare the performance of the two phases, a normalization formula was applied to adapt the different maximum times for each subject and for each speech. This process was necessary to ensure a fair and accurate evaluation of relative performance. The formula used to calculate the ratio was as follows:

$$Ps_{RTs\text{ratio}} = \frac{Ps_{RTs\text{ subject}}}{Ps_{RTs\text{ max}}}$$

$$Ss_{RTs\text{ratio}} = \frac{Ss_{RTs\text{ subject}}}{Ss_{RTs\text{ max}}}$$

This formula provides a standardized measure of the relative performance of the preparation and speech stage, taking into account the different maximum times for each stage.

In addition, objective and qualitative criteria were employed to assess the suitability, cohesion of content, and congruence of each

speech. Interventions that did not meet the relevance criteria, characterized by a lack of substance, inclusion of off-topic arguments, or inconsistency with the task, were excluded from analysis. The criteria used include: (i) relevance (the ability of each discourse to address the proposed topics in a significant and appropriate manner); (ii) coherence (logical and structural coherence of content, ensuring a uniform and comprehensible flow of thought); (iii) appropriateness (conformity to standards and expectations related to the context and objectives of the analysis); (iv) productivity (the ability to provide substantial and relevant contributions, avoiding superfluous or irrelevant arguments). This process ensured the equivalence of content among the five-discourse analyzed, facilitating an objectively comparable evaluation of performance. Analysis did not include discourses that did not match productivity criteria, such as those that were incoherent, lacked content, or included off-topic or inconsistent subjects (rejected discourses, 0.769 %).

2.5. EEG data acquisition and processing

For EEG data collection, an 18-channel DC amplifier (SYNAMPs) and NEUROSCAN 4.2 software were utilized. ElectroCap with Ag/AgCl electrodes was positioned on 18 scalp sites according to the 10/20 system, with reference to the earlobes. Additionally, two electrooculographic electrodes were placed around each participant’s left eye to minimize visual interference. Data were sampled at 1000 Hz, filtered with a 50 Hz notch input filter, and electrode impedance was monitored to ensure it remained below 5 kΩ. Subsequently, offline processing of data from both eyes-open and eyes-closed resting states and related to the Pp₁₋₅ and the Ep₁₋₅ was carried out using Vision Analyzer2 software (Brain Products GmbH, Gilching, Germany), including IIR bandpass filtering (0.1–50 Hz, 48 dB/octave) and correction with an ICA-based algorithm (Jung et al., 2000). After electrooculographic artifact correction and thorough visual inspection, only segments devoid of muscle, eye artifacts, or other disturbances were retained. Statistical analysis was conducted using data from all electrodes. The data were epoched with a 2000 ms window to maintain EEG data integrity, and artifact-free segments were used to compute condition-specific power density spectra (PD) through fast Fourier transform (Hamming window, resolution: 0.5 Hz). Average PD for delta (0.5–3.5 Hz), theta (4–7.5 Hz), alpha (8–12.5 Hz), beta (13–30 Hz), and gamma (30.5–50 Hz) frequency bands was then extracted for each phase. Additionally, task-related data were normalized to each participant’s eyes-open baseline.

2.6. Autonomic data acquisition and processing

Cardiovascular (heart rate – HR, heart rate variability – HRV, inter-beat interval – IBI, blood volume pulse – BVP, and pulse volume amplitude – PVA) and electrodermal activity (skin conductance level – SCL, and skin conductance response – SCR) data were recorded using a wireless Biofeedback2000Xpert system with a MULTI radio module (Schuhfried GmbH, Modling, Austria). The sensor was placed on the second finger of the non-dominant hand, and hand movements were tracked using an accelerometer to minimize interference. Data were sampled at 40 Hz.

After qualitative and quantitative inspection to remove artifacts, autonomic activity during baseline and task phases was segmented and averaged to calculate mean condition specific SCL, SCR, BVP, PVA, and HR modulations. IBI metrics were derived from raw HR data, while HRV was computed as the standard deviation of IBI to assess vagal influence on cardiovascular activity. Similar to EEG data, also task-related autonomic data were normalized to each participant’s eyes-open baseline.

2.7. Statistical analysis

Repeated measures ANOVAs (analyses of variance) (Tabachnick & Fidell, 2007) were applied to behavioural, EEG, and autonomic data

using IBM SPSS (version 27, IBM Corp., Chicago, IL, USA).

For behavioural SST data, a repeated measures ANOVA with *Stage* (2: Ps, Ss) and *Discourses* (5) as the within-subject factor was applied to the behavioural RTs scores.

For EEG data, five repeated measures ANOVAs considering *Stage* (2: Ps, Ss), *Discourses* (5) and *Electrode* (18) as within-subject independent variables were carried out for the five different frequency bands (delta, theta, alpha, beta, and gamma).

Finally, for autonomic data, six repeated measures ANOVAs with *Stage* (2: Ps, Ss) and *Discourses* (5) as within-subject independent variable were applied to the following autonomic indices considered as dependent variables: SCL, SCR, BVP, PVA, HR, and HRV.

When necessary, degrees of freedom for each ANOVA test were adjusted using the Greenhouse-Geisser epsilon. After that, significant interactions were investigated using pairwise comparisons to assess simple effects, and potential biases arising from multiple comparisons were addressed by using the Bonferroni correction. Eta squared (η^2) indices were used to measure the size of statistically significant effects, with a significance level of $\alpha = 0.05$. The normal distribution of the data was assessed by using skewness and kurtosis test in a preliminary statistical phase.

3. Results

3.1. Behavioural results

Considering the analysis performed on the behavioural SST score, a significant main effect was found for *Stage* ($F_{(1, 25)} = 141.92, p \leq 0.001, \eta^2 = 0.850$) with higher mean for the RTs-ratio in Ss compared to the Ps (Fig. 2a). Furthermore, a significant main effect was observed for *Discourses* ($F_{(2.42, 60.5)} = 9.857, p \leq 0.001, \eta^2 = 0.283$) (Fig. 2b). Significant pairwise comparisons revealed a higher RTs-ratio in D1 compared to D2 ($p = 0.002$) and D5 ($p \leq 0.001$). No other significant differences were found in the behavioural data.

3.2. EEG results

The results from the ANOVA tests performed on the different frequency bands (delta, theta, alpha, beta, and gamma) are reported below.

3.2.1. Delta band

For delta band, a significant main effect was found for *Stage* ($F_{(1, 15)} = 60.991, p \leq 0.001, \eta^2 = 0.803$), with higher mean values of delta

power in the Ss compared to the Ps.

Furthermore, an interaction effect was revealed for *Stage* $\times$ *Electrode* ($F_{(17, 255)} = 4.249, p \leq 0.001, \eta^2 = 0.221$) with higher mean values of delta power in the Ep compared to the Ps for all the electrode considered.

Also, an interaction effect was found for *Stage* $\times$ *Discourses* $\times$ *Electrode* ($F_{(68, 1020)} = 11.457, p \leq 0.001, \eta^2 = 0.433$) (Fig. 3). Firstly, in the D1, significant higher delta band values were found for all electrodes in the Ss compared to the Ps, except for Fp1, T7, C4, P3, Pz, and O1. In particular, during D1, significant lower delta band values were found for T7 in the Ss compared to the Ps. Secondly, in the D2, significant higher delta band values were found for all electrodes in the Ss compared to the Ps, except for AFz, Fp1, Fp2, and F7.

3.2.2. Theta band

For theta band, a significant main effect was found for *Stage* ($F_{(1, 15)} = 28.348, p \leq 0.001, \eta^2 = 0.654$), with higher mean values of theta power in the Ss compared to the Ps.

Likewise, an interaction effect was revealed for *Stage* $\times$ *Electrode* ($F_{(17, 255)} = 4.454, p \leq 0.001, \eta^2 = 0.229$) with higher mean values of theta power in the Ss compared to the Ps for all the electrode considered, with the exception of electrode P4.

Moreover, an interaction effect was found for *Stage* $\times$ *Discourses* $\times$ *Electrode* ($F_{(68, 1020)} = 13.110, p \leq 0.001, \eta^2 = 0.466$) (Fig. 4). Firstly, in the D1, significant higher theta band values were found for all electrodes in the Ss compared to the Ps, except for Fp1, F4, T7, T8, and O1. In particular, during D1, significant lower theta band values were found for T7 in the Ss compared to the Ps. Secondly, in the D2, significant higher theta band values were found for all electrodes in the Ss compared to the Ps, except for AFz, Fp1, Fp2, and P4. In the D3, the same trend was confirmed except for P4, as well as for AFz and Fp1 in D4. Finally, in the D5, significant higher theta band values were found for all electrodes in the Ss compared to the Ps, except for C4, P3, Pz, and P4.

3.2.3. Alpha band

For alpha band, a significant main effect was found for *Stage* ($F_{(1, 15)} = 26.935, p \leq 0.001, \eta^2 = 0.642$), with higher mean values of alpha power in the Ss compared to the Ps.

An interaction effect was also revealed for *Stage* $\times$ *Electrode* ($F_{(17, 255)} = 4.454, p \leq 0.001, \eta^2 = 0.229$) with higher mean values of alpha power in the Ss compared to the Ps for all the electrode considered.

Furthermore, an interaction effect was found for *Stage* $\times$ *Discourses* $\times$ *Electrode* ($F_{(68, 1020)} = 5.651, p \leq 0.001, \eta^2 = 0.274$) (Fig. 5). Specifically,

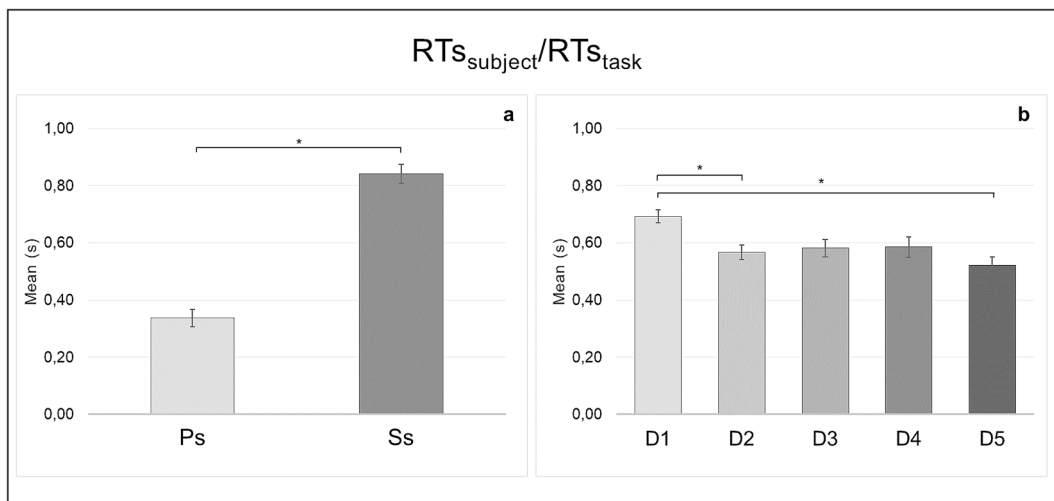


Fig. 2. a-b. Behavioural SST results. (a) The bar graph shows statistically significant differences in RTs ratio scores for the Ps and the Ss. (b) The bar graph shows statistically significant differences in RTs ratio scores for each discourse. For all graph, bars represent ± 1 Standard Error and stars (*) mark statistically significant comparisons.

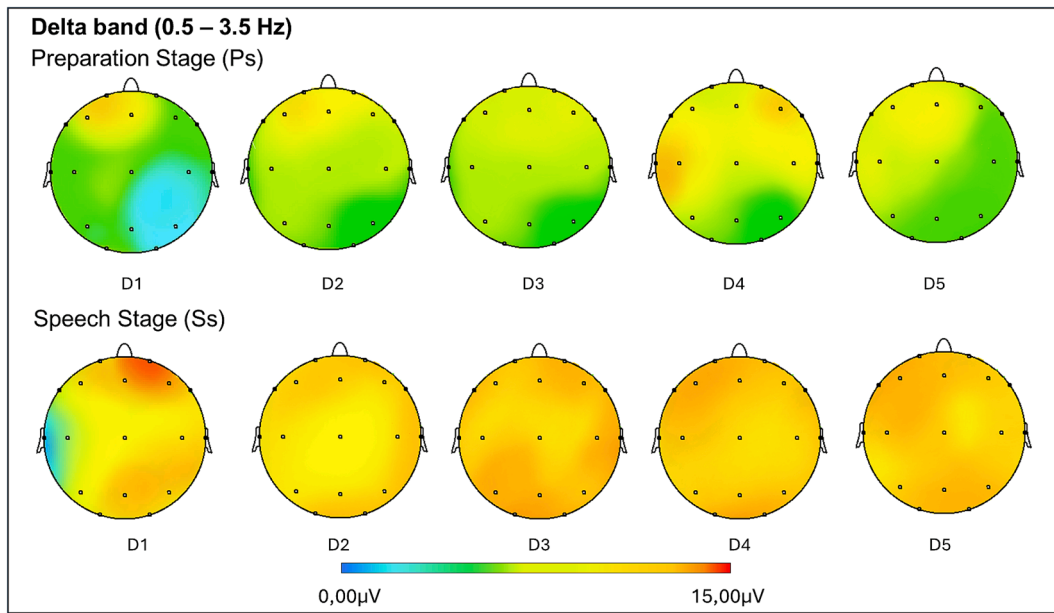


Fig. 3. EEG Delta results. The topographic EEG maps of spectral power density of the interaction effects in *Stage* $\times$ *Discourses* $\times$ *Electrode* for the delta band is reported. For topographic EEG maps, red represented an increase in power for delta band (software: Vision Analyser2).

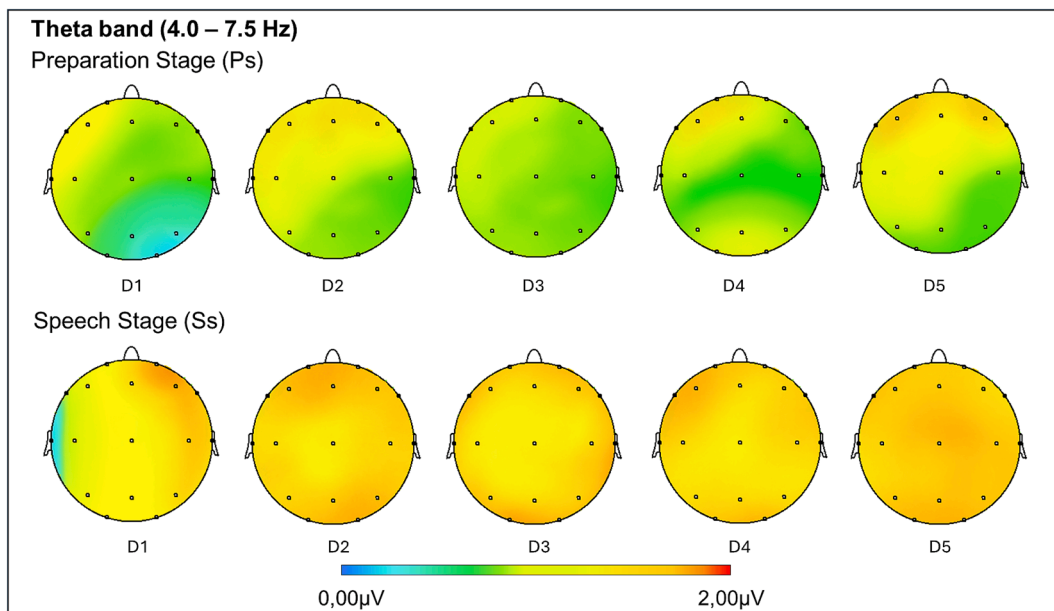


Fig. 4. EEG Theta results. The topographic EEG maps of spectral power density of the interaction effects in *Stage* $\times$ *Discourses* $\times$ *Electrode* for the theta band is reported. For topographic EEG maps, red represented an increase in power for theta band (software: Vision Analyser2).

in the D1, significant higher alpha band values were found for all electrodes in the Ss compared to the Ps, except for F4, F8, T7, T8, P4, and O1. Secondly, in the D2, significant higher alpha band values were found for all electrodes in the Ss compared to the Ps, except for C3, Cz, C4, P3, and Pz. In the D3, higher mean values of alpha power in the Ss compared to the Ps for all the electrode considered. While in the D4, higher mean values of alpha power in the Ss compared to the Ps for all electrodes, except for AFz, C3, Cz, C4, P3 and Pz. Finally, in the D5, significant higher alpha band values were found for all electrodes in the Ss compared to the Ps, except for Fp1, Fz, C3, Cz, C4, P3, Pz and P4.

3.2.4. Beta band

For beta band, a significant main effect was found for *Stage* ($F_{(1, 15)} =$

$20.732, p \leq 0.001, \eta^2 = 0.580$), with higher mean values of beta power in the Ss compared to the Ps.

Moreover, an interaction effect was revealed for *Stage* $\times$ *Electrode* ($F_{(17, 255)} = 9.139, p \leq 0.001, \eta^2 = 0.379$) with higher mean values of beta power in the Ss compared to the Ps for all the electrode considered.

Also, an interaction effect was found for *Stage* $\times$ *Discourses* $\times$ *Electrode* ($F_{(68, 1020)} = 1.488, p = 0.008, \eta^2 = 0.090$) (Fig. 6). Specifically, in the D1, D2 and D3, significant higher beta band values were found for all electrodes in the Ss compared to the Ps, except for P4 in the D1. In the D4, higher mean values of beta power in the Ss compared to the Ps for all electrodes, except for AFz, Fp1, Fp2, F7, and C4. Finally, in the D5, significant higher beta band values were found for all electrodes in the Ss compared to the Ps, except for Fp1, Fz, Fp2, C3, Cz, C4, P3, Pz and P4.

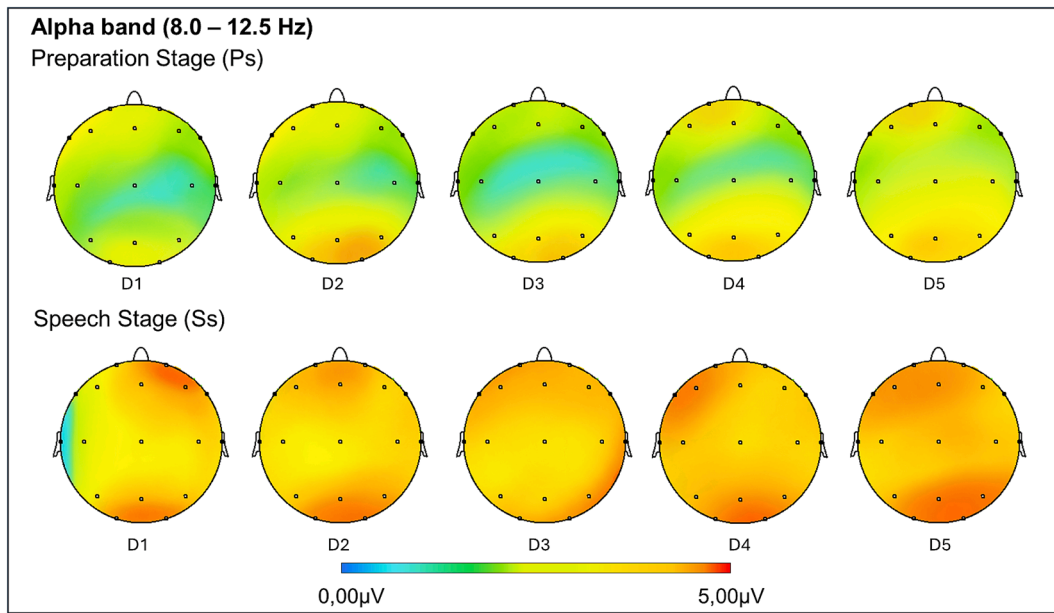


Fig. 5. EEG Alpha results. The topographic EEG maps of spectral power density of the interaction effects in *Stage* $\times$ *Discourses* $\times$ *Electrode* for the alpha band is reported. For topographic EEG maps, red represented an increase in power for alpha band (software: Vision Analyser2).

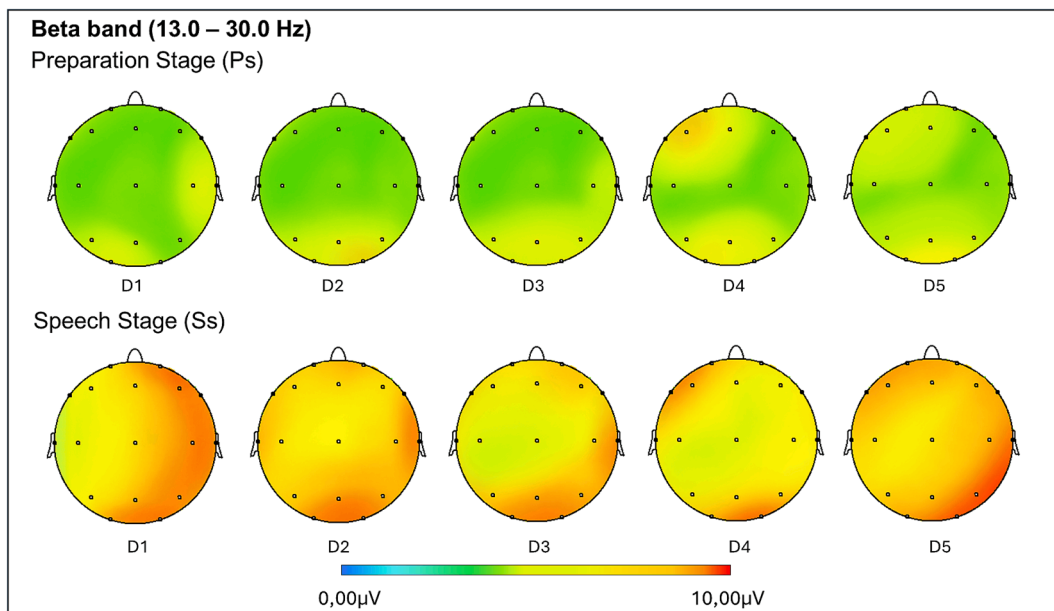


Fig. 6. EEG Beta results. The topographic EEG maps of spectral power density of the interaction effects in *Stage* $\times$ *Discourses* $\times$ *Electrode* for the beta band is reported. For topographic EEG maps, red represented an increase in power for beta band (software: Vision Analyser2).

3.2.5. Gamma band

For gamma band, a significant main effect was found for *Stage* ($F_{(1, 15)} = 9.433$, $p = 0.008$, $\eta^2 = 0.386$), with higher mean values of gamma power in the Ss compared to the Ps.

An interaction effect was revealed for *Stage* $\times$ *Discourses* ($F_{(2.59, 33.84)} = 3.248$, $p = 0.038$, $\eta^2 = 0.178$). Significant pairwise comparisons revealed higher mean values of gamma power in D₂ compared to D₁ ($p = 0.033$) during the Ss, and higher mean values for the Ss compared to the Ps in D₁ ($p = 0.008$), D₂ ($p = 0.010$), D₃ ($p = 0.007$), D₄ ($p = 0.021$), and D₅ ($p = 0.031$). An interaction effect was also found for *Stage* $\times$ *Electrode* ($F_{(17, 255)} = 4.902$, $p \leq 0.001$, $\eta^2 = 0.246$) with higher mean values of gamma power in the Ss compared to the Ps for all the electrode considered.

Moreover, an interaction effect was found for *Stage* $\times$ *Discourses* $\times$ *Electrode* ($F_{(68, 1020)} = 1.873$, $p \leq 0.001$, $\eta^2 = 0.111$) (Fig. 7). Firstly, in the D₁, significant higher gamma band values were found for all electrodes in the Ss compared to the Ps, except for AFz, Fp1, Fz and P4. Secondly, in the D₂ and D₃, significant higher gamma band values were found for all electrodes in the Ss compared to the Ps, except for O2 in the D₂, and F3 and O2 in the D₃. Thirdly, in the D₄, significant higher gamma band values were found for all electrodes in the Ss compared to the Ps, excluding AFz, Fp1, F7, F3, Fz, C3, and O2. Finally, in the D₅, significant higher gamma band values were found for all electrodes in the Ss compared to the Ps, excluding Fp1, Fp2, Fz, C3, Cz, C4, P3, Pz and P4.

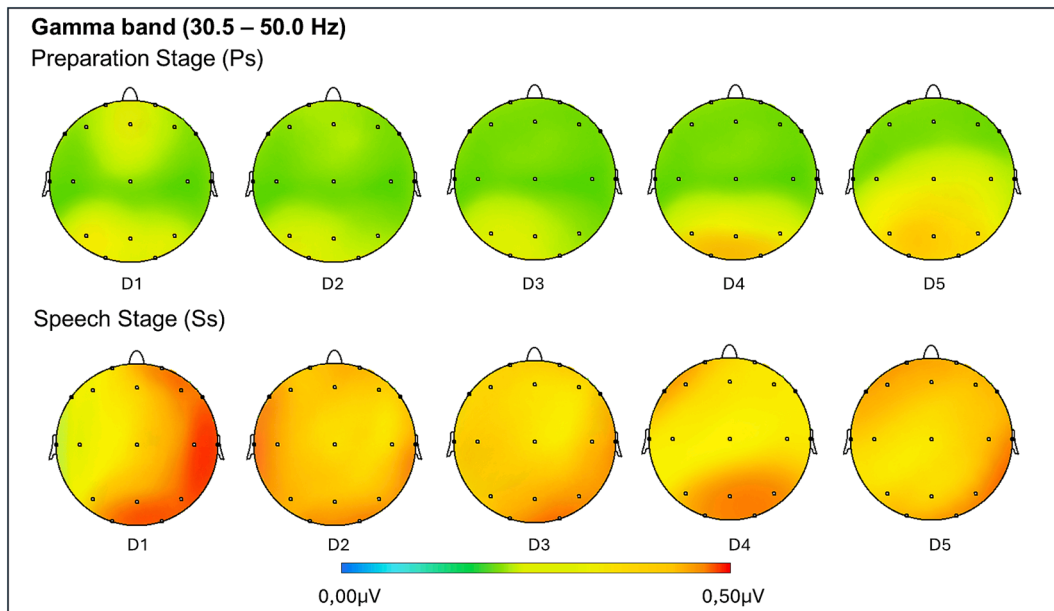


Fig. 7. EEG Gamma results. The topographic EEG maps of spectral power density of the interaction effects in *Stage* $\times$ *Discourses* $\times$ *Electrode* for the gamma band is reported. For topographic EEG maps, red represented an increase in power for gamma band (software: Vision Analyser2).

3.3. Autonomic results

For PVA, a significant main effect was revealed for *Stage* ($F_{(1, 25)} = 4.357, p = 0.047, \eta^2 = 0.148$), with higher mean values for the Ps compared to the Ss. Furthermore, an interaction effect was found for *Stage* $\times$ *Discourses* ($F_{(2.24, 56.01)} = 8.889, p \leq 0.001, \eta^2 = 0.262$) (Fig. 8a). Significant pairwise comparisons revealed higher mean values in the Ss compared to the Ps ($p = 0.039$) in D1. On the contrary, lower mean values in the Ss compared to the Ps were found in D2 ($p = 0.035$), D3 ($p = 0.032$), D4 ($p = 0.033$), and D5 ($p = 0.046$). Significant pairwise comparisons revealed also higher mean value of PVA in D1 compared to D2 ($p = 0.011$), D4 ($p = 0.033$), and D5 ($p = 0.043$) during the Ps.

For HRV, a significant main effect was revealed for *Stage* ($F_{(1, 25)} = 12.381, p = 0.002, \eta^2 = 0.331$) (Fig. 8b), with higher mean values for the Ss compared to the Ps.

No other significant results were found for HR, BVP, SCL and SCR.

4. Discussion

The current study investigates the behavioral, EEG, and autonomic responses to stress during the preparation and speech stages of five different discourses presented in an SST among healthy adults. Findings showed an increase of cognitive resources needed for facing the speech stage compared to the preparation stage, as highlighted by the higher RTs, the cardiovascular response of PVA, HRV and the generalized increase of EEG delta, theta, alpha, beta, and gamma frequency bands in this specific stage. Interestingly, the increase of emotional salience produced by the five discourses induced a negativity bias (reduction of the RTs for) and a whole scalp activation for the low frequency bands (mirroring the emotional processing) accompanied by a progressively lower cognitive effort (shown by both high-frequency bands and autonomic indices) for the most emotionally stressful conditions (the last and most challenging discourses).

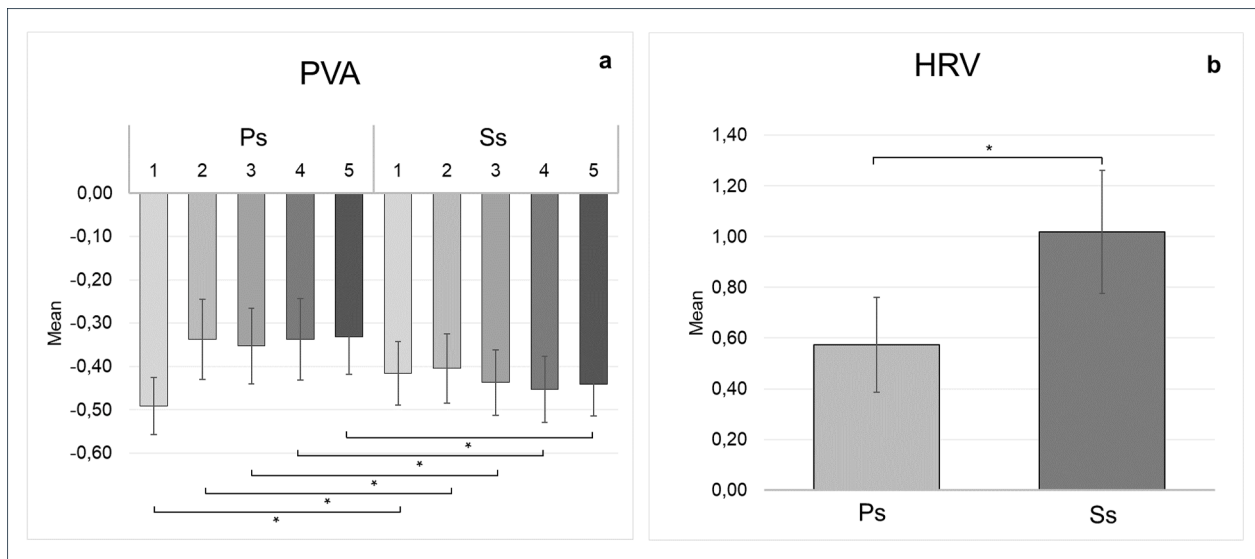


Fig. 8. a-b. Autonomic results. (a) The bar graph shows significant differences for PVA for each Stage and discourses. (b) The bar graph shows significant differences for HRV for the Ps and the Ss. For all graph, bars represented ± 1 Standard Error and stars (*) mark statistically significant comparisons.

In line with our first hypothesis on the behavioral data, we found an increase in RTs during the speech stage (Ss) compared to the preparation stage. As a cognitive metric, the longer RTs required for handling stress during the speech stage indicate the need for additional cognitive resources to access stored information efficiently, serving as an index of fluency and coherence in language production (Huang & Shanks, 2021). As expected, participants seem to engage more cognitive resources during this critical moment.

Interestingly, when considering the five discourses, we observed a decrease in RTs in D5 and D2 compared to D1. D1 required participants to prepare the best presentation of themselves, and the social feedback they received from the judges was delivered through a friendly video. This increase in time for D1 can be explained by the greater investment of cognitive resources due to its being the first speech and video. Conversely, D2 was the first instance where participants faced a more challenging situation, requiring them to present a scenario where they encountered difficulty in making a decision and received social feedback of neutrality and disregard from the judges. D5 was the most challenging discourse, as participants had to describe a situation where they made a decision in disagreement with the rest of the group, and the judges expressed complete aversion both facially and verbally. This decrease in RTs for D2 and D5 can be explained by the exposure to highly emotionally salient stimuli, which may induce an attentional bias corresponding to increased emotional salience (i.e., negativity bias; (Balconi & Pozzoli, 2009). Thus, the emotional escalation associated with these discourses could lead to reduced RTs.

Secondly, regarding the EEG results, we found a generalized increase of low frequency bands (delta and theta band) during the speech compared to the preparation stage. This result confirmed our hypothesis for which the speech stage can be considered the inner and most stressful stage of the interview compared to the preparation stage, requiring higher emotional processing (Barzegar et al., 2023). In this previous study (Barzegar et al., 2023), the increase of delta band frequency has been related to an internal focus of the participant to find the correct answer and avoid shame in front of examiners during the TSST test, which is considered a mental activity with a negative feeling.

Moreover, during the speech compared to the preparation stage, the observed increase of theta band could be due to the exposure to the social feedback (Yao et al., 2020), as well as to the emotional adaptation processes and stress regulation (Güntekin & Başar, 2016; Marox et al., 2001). Thus, in our study, the observed increase in low-frequency bands during the speech stage, compared to the preparation stage, might indicate the processing of an emotional situation perceived as stressful, challenging, and potentially carrying a negative connotation.

In line with the previously discussed evidence on low frequency bands, we also observed a generalized increase of high frequency bands during speech compared to the preparation stage. In the literature, high-frequency EEG oscillations (alpha, beta, and gamma) are typically linked to brain states that demand attention and alertness (Vanhollebeke et al., 2022). Specifically, the alpha band is believed to be inversely correlated with cortical activity and reflecting coordination mechanisms of brain networks (Allen, Coan, & Nazarian, 2004; Jensen & Mazaheri, 2010; Mathewson et al., 2011), the beta band is associated with attention and sustaining the current cognitive or sensorimotor state (Engel & Fries, 2010; Wróbel, 2000), while gamma oscillations were linked to cognitive as well as learning processes (Fitzgibbon, Pope, MacKenzie, Clark, & Willoughby, 2004; Miltner, Braun, Arnold, Witte, & Taub, 1999).

Former works found an increase of alpha power in the reactive phase compared to the baseline (Betti et al., 2018) and reported the significantly higher beta power in frontal electrodes during the reactive phase (Hafeez et al., 2018; Minguillon et al., 2016; Perrin et al., 2019). This evidence suggests that the increase in high frequency bands activation marks the most stressful phase, that is the reactive phase within the stress response curve. Therefore, entering more specifically into the stages that could characterize the reactive phase, this result suggests

that, between the preparation and the speech stage, the latter – which coincides with the behavioral and verbal performance – appears more stressful and requires greater cognitive effort.

Single electrode results regarding the delta and theta bands show how the increase in emotional requests and more negative feedback require the involvement of an activation that concerns almost the whole scalp, always with greater activation for the speech compared to the preparation stage. Interestingly, it is possible to notice how in the left temporal area there is a greater presence of both the delta band and the theta band for preparation, compared to the speech stage for the first discourse (D1). In a former work, it was found that acute stress increases memory-related theta oscillations in the medial temporal lobe, potentially enhancing the storage of emotionally significant negative events during stressful situations (Heinbockel, Quaedflieg, Schneider, Engel, & Schwabe, 2021). In our study, this effect of increased activity in the temporal area is localized on the left for the delta and theta bands for D1 during the preparation stage (the first stressful request and before any feedback), suggesting that when faced with a stressful task, individuals might prepare by retrieving emotional information from memory.

On the other hand, single electrode results for alpha, beta, and gamma oscillations show a sort of plateau effect from a cognitive point of view, as if, with an increase in speeches, there seems to be less involvement of the entire scalp in the speech stage. There could be two possible interpretations for this effect. First, a possible interpretation could be that this effect may be due to habituation mechanisms to repeated stress exposure, for which progressively less cognitive effort and neural resources are required by the participants. Alternatively, this plateau effect could be explained by the development of coping skills by participants who, with the increase in the emotional salience of the discourses – despite a high emotional activation as shown by the result of the delta and theta bands – become more able to manage the speech stage. Future studies should be conducted to validate these considerations on EEG data by conducting analyses on both single electrodes and specific regions of interest.

Finally, autonomic results displayed significant variations for the different discourses in the PVA index, which according to the literature, provides information about individuals' emotional arousal and cognitive effort (MacPherson, Abur, & Stepp, 2017). During the preparation stage, PVA levels were notably higher for the first discourse (D1) compared to the others (D2, D4, and D5). This increase may be attributed to the substantial cognitive effort required for participants to create an effective self-presentation. Since preparing to present oneself favorably involves intense self-reflection and planning, it might naturally demand more cognitive resources than subsequent tasks.

The speech stage further illustrated such distinct impact of the first discourse on individuals' cardiovascular response: indeed, higher values of PVA were found for the speech compared to the preparation stage for the D1, suggesting that delivering this initial self-presentation was particularly engaging. In contrast, the lower PVA values during the speech stage for D2, D3, D4, and D5 compared to their preparation stages indicate a different pattern. In these discourses, participants might have experienced reduced cognitive load during the speeches. This result diverges from the expected trend of increasing cardiovascular indices with heightened emotional salience in later discourses, which were characterized by more challenging demands and negative feedback. One plausible explanation for these findings is the unique nature of the first discourse. As first challenge, it likely elicited the highest level of emotional activation and cognitive engagement. Additionally, the positive social feedback received from the evaluation board during D1 could have further intensified participants' arousal and effort, as social validation is a powerful motivator and stressor. In contrast, the subsequent discourses, despite being more demanding, might not have matched the emotional impact of the first discourse, as participants had already acclimated to the evaluation context and the novelty gradually decreased. Or again, the two possible interpretations mentioned for the EEG high-frequency bands results could explain this lower PVA values

during the speech stage for D2, D3, D4, and D5 compared to their preparation stages. On the one hand, this effect could be due to habituation mechanisms derived from repeated stress exposure, which require less and less neural resources to be addressed; on the other hand, the development of coping skills allows participants to better face the speech stage discourse by discourse.

In addition, we observed a general increase in HRV during the speech compared to the preparation stage. In the literature, HRV has been systematically associated to the intensity of stress response and to the efficiency of coping mechanisms (Appelhans & Luecken, 2006; Obrist, 1981; Thayer, Åhs, Fredrikson, Sollers, & Wager, 2012). This evidence may suggest that despite the increase of individuals' emotional arousal and cognitive effort experienced during the speech stage, our group of participants seems to demonstrate to be able to cope with this source of stress.

Overall, this study is not exempt from limitations. First, the relatively small sample size may limit the generalizability of the findings and future studies could include larger and more diverse groups of participants. Secondly, while the SST simulates a real-life interview scenario, the controlled laboratory setting may not entirely replicate the complexities of real-world stressors. Thirdly, despite EEG can offer deeper insights into the neural mechanisms underlying stress and emotional regulation, to grasp the complexity of stress response this study could benefit from the integration of additional physiological and psychological measures, such as hormonal analysis (e.g., cortisol levels), or self-report questionnaires on perceived stress and coping strategies, as well as neuroimaging techniques such as the functional Near Infrared Spectroscopy for a more detailed localization of the neural activation. Moreover, we manipulated the exposure to progressively negative emotional stimuli, which might not reflect the experience in real interviews (in which there may be a plateau phase and habituation to stress levels due to coping strategies). Finally, to the best of our knowledge, no previous studies distinguished between the preparation and speech stages and considered the emotional and cognitive processes related to these stages, thus data interpretation must be taken with caution.

Future studies should include larger and more diverse participant groups to enhance the validity and generalizability of the findings. This would allow for the examination of individual differences in stress responses and coping mechanisms. Also, this study could be expanded to real-world settings and can provide insights into how people manage stress in everyday environments or be exploited to enhance public speaking skills and emotion regulation in healthy samples. Moreover, to investigate the efficacy of stress management interventions (e.g., mindfulness, cognitive-behavioral strategies) implemented before or after the stressor could provide practical applications for improving stress resilience. Comparing these interventions across different phases of the stress response could identify optimal timing and methods for stress reduction.

Ethics statement.

The study's experimental design and procedures were reviewed and approved by the Ethics Committee of the Department of Psychology at the Catholic University of the Sacred Heart, Milan, Italy, and according to the GDPR – Reg. UE 2016/679 and its ethical guidelines. The participants provided their written informed consent to participate in this study.

CRedit authorship contribution statement

Laura Angioletti: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Katia Rovelli:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Michela Balconi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported by Fondazione Cariplo, grant number 2023-1612 (Progetto “Empowering Emotions regulation and Executive functions in Behavioral Addictions (E3BA)”, ID. 2023-1612 - Fondazione Cariplo).

Data availability

Data will be made available on request.

References

- Allen, J. B., Coan, J. A., & Nazarian, M. (2004). Issues and assumptions on the road from raw signals to metrics of frontal EEG asymmetry in emotion. *Biol Psychol*, 67, 183–218. <https://doi.org/10.1016/J.BIOPSYCHO.2004.03.007>
- Allen, A. P., Kennedy, P. J., Dockray, S., Cryan, J. F., Dinan, T. G., & Clarke, G. (2017). The Trier Social Stress Test: Principles and practice. *Neurobiol Stress*, 6, 113. <https://doi.org/10.1016/J.YNSTR.2016.11.001>
- F. Al-Shargie, M. Kiguchi, N. Badruddin, S.C. Dass, A. Fadzil Mohammad Hani, T. Boon Tang, B. Czéh, T. Michaelis, T. Watanabe, J. Frahm, G. de Biurrun, M. van Kampen, A. Bartolomucci, E. Fuchs. Mental stress assessment using simultaneous measurement of EEG and fNIRS. *Biomedical Optics Express*, 7, Issue 10. 3882–3898 7 (2016) 3882–3898. <https://doi.org/10.1364/BOE.7.003882>
- F. Al-Shargie, B.T. Tang, M. Kiguchi, S. Bishop, J. Duncan, M. Brett, A.D. Lawrence. Assessment of mental stress effects on prefrontal cortical activities using canonical correlation analysis: an fNIRS-EEG study. *Biomedical Optics Express*, 8, Issue 5. 2583–2598 8 (2017) 2583–2598. <https://doi.org/10.1364/BOE.8.002583>
- Al-Shargie, F., Tang, B. T., Badruddin, N., & Kiguchi, M. (2018). Towards multilevel mental stress assessment using SVM with ECoG: An EEG approach. *Med Biol Eng Comput*, 56, 125–136. <https://doi.org/10.1007/S11517-017-1733-8/FIGURES/6>
- Al-Shargie, F., Tang, B. T., & Kiguchi, M. (2017b). Stress Assessment Based on Decision Fusion of EEG and fNIRS Signals. *IEEE Access*, 5, 19889–19896. <https://doi.org/10.1109/ACCESS.2017.2754325>
- Appelhans, B. M., & Luecken, L. J. (2006). Heart Rate Variability as an Index of Regulated Emotional Responding. <https://doi.org/10.1037/1089-2680.10.3.229>
- Balconi, M., Grippa, E., & Vanutelli, M. E. (2015). What hemodynamic (fNIRS), electrophysiological (EEG) and autonomic integrated measures can tell us about emotional processing. *Brain Cogn*, 95, 67–76. <https://doi.org/10.1016/J.BANDC.2015.02.001>
- Balconi, M., & Pozzoli, U. (2009). Arousal effect on emotional face comprehension: Frequency band changes in different time intervals. *Physiol Behav*, 97, 455–462. <https://doi.org/10.1016/J.PHYSBEH.2009.03.023>
- Barzegar, M., Jahromi, G. P., Meftahi, G. H., & Hatf, B. (2023). The Complexity of Electroencephalographic Signal Decreases during the Social Stress. *J Med Signals Sens*, 13, 57. https://doi.org/10.4103/JMSS.JMSS_131_21
- Betti, S., Lova, R. M., Rovini, E., Acerbi, G., Santarelli, L., Cabiati, M., Del Ry, S., & Cavallo, F. (2018). Evaluation of an Integrated System of Wearable Physiological Sensors for Stress Monitoring in Working Environments by Using Biological Markers. *IEEE Trans Biomed Eng*, 65, 1748–1758. <https://doi.org/10.1109/TBME.2017.2764507>
- Boesch, M., Sefidan, S., Ehlert, U., Annen, H., Wyss, T., Steptoe, A., & La Marca, R. (2014). Mood and autonomic responses to repeated exposure to the Trier Social Stress Test for Groups (TSST-G). *Psychoneuroendocrinology*, 43, 41–51. <https://doi.org/10.1016/J.PSYNEUEN.2014.02.003>
- Campbell, J., & Ehlert, U. (2012). Acute psychosocial stress: Does the emotional stress response correspond with physiological responses? *Psychoneuroendocrinology*, 37, 1111–1134. <https://doi.org/10.1016/J.PSYNEUEN.2011.12.010>
- Cohen, S., Kamarck, T., & Mermelstein, R. (1983). A global measure of perceived stress. *J Health Soc Behav*, 385–396.
- Dickerson, S. S., & Kemeny, M. E. (2004). Acute stressors and cortisol responses: A theoretical integration and synthesis of laboratory research. *Psychol Bull*, 130, 355–391. <https://doi.org/10.1037/0033-2909.130.3.355>
- J.W.B. Eelsey, A.I. Filmer, H.R. Galvin, J.D. Kurath, L. Vossoughi, L.S. Thomander, M. Zavodnik, M. Kindt. Reconsolidation-based treatment for fear of public speaking: a systematic pilot study using propranolol. *Translational Psychiatry*. 2020 10:1 10 (2020) 1–12. <https://doi.org/10.1038/s41398-020-0857-z>
- Engel, A. K., & Fries, P. (2010). Beta-band oscillations — signalling the status quo? *Curr Opin Neurobiol*, 20, 156–165. <https://doi.org/10.1016/j.conb.2010.02.015>
- Engert, V., Efanov, S. I., Duchesne, A., Vogel, S., Corbo, V., & Pruessner, J. C. (2013). Differentiating anticipatory from reactive cortisol responses to psychosocial stress. *Psychoneuroendocrinology*, 38, 1328–1337. <https://doi.org/10.1016/J.PSYNEUEN.2012.11.018>

- Fitzgibbon, S. P., Pope, K. J., MacKenzie, L., Clark, C. R., & Willoughby, J. O. (2004). Cognitive tasks augment gamma EEG power. *Clinical Neurophysiology*, 115, 1802–1809. <https://doi.org/10.1016/j.clinph.2004.03.009>
- Güntekin, B., & Başar, E. (2016). Review of evoked and event-related delta responses in the human brain. *International Journal of Psychophysiology*, 103, 43–52. <https://doi.org/10.1016/j.ijpsycho.2015.02.001>
- M.A. Hafeez, S. Shakil, S. Jangsher. Stress Effects on Exam Performance using EEG, 2018 14th International Conference on Emerging Technologies. ICET. 2018. (2019). <https://doi.org/10.1109/ICET.2018.8603652>.
- Heinbockel, H., Quaedflieg, C. W. E. M., Schneider, T. R., Engel, A. K., & Schwabe, L. (2021). Stress enhances emotional memory-related theta oscillations in the medial temporal lobe. *Neurobiol Stress*, 15. <https://doi.org/10.1016/J.YNSTR.2021.100383>
- Hellhammer, J., & Schubert, M. (2012). The physiological response to Trier Social Stress Test relates to subjective measures of stress during but not before or after the test. *Psychoneuroendocrinology*, 37, 119–124. <https://doi.org/10.1016/J.PSYNEUEN.2011.05.012>
- Huang, T. S. T., & Shanks, D. R. (2021). Examining the relationship between processing fluency and memory for source information. *R Soc Open Sci*, 8. <https://doi.org/10.1098/RSOS.190430>
- Izhar, L. I., Roslan, N. S., Feng, Y. X., Faye, I., Ho, E. T. W., & Rahman, M. A. (2019). Classification of neuroticism using psychophysiological signals during speaking task based on two different baseline measurements, *International Journal of Integrated Engineering*, 11, 119–128. <https://doi.org/10.30880/IJIE.2019.11.03.013>
- Jensen, O., & Mazaheri, A. (2010). Shaping functional architecture by oscillatory alpha activity: Gating by inhibition. *Front Hum Neurosci*, 4, 2008. <https://doi.org/10.3389/FNHUM.2010.00186/BIBTEX>
- Jung, T.-P., Makeig, S., Humphries, C., Lee, T.-W., McKeown, M. J., Iragui, V., & Sejnowski, T. J. (2000). Removing electroencephalographic artifacts by blind source separation. *Psychophysiology*, 37, 163–178. <https://doi.org/10.1111/1469-8986.3720163>
- Kirschbaum, C., Pirke, K. M., & Hellhammer, D. H. (1993). The ‘Trier Social Stress Test’—a tool for investigating psychobiological stress responses in a laboratory setting. *Neuropsychobiology*, 28, 76–81. <https://doi.org/10.1159/000119004>
- MacPherson, M. K., Abur, D., & Stepp, C. E. (2017). Acoustic Measures of Voice and Physiologic Measures of Autonomic Arousal during Speech as a Function of Cognitive Load. *J Voice*, 31(504), e1–504.e9. <https://doi.org/10.1016/J.JVOICE.2016.10.021>
- Marox, E., Rodríguez, H., Yañez, G., Bernal, J., Rodríguez, M., Fernández, T., Silva, J., Reyes, A., & Guerrero, V. (2001). Broad band spectral measurements of EEG during emotional tasks. *International Journal of Neuroscience*, 108, 251–279.
- Mathewson, K. E., Lleras, A., Beck, D. M., Fabiani, M., Ro, T., & Gratton, G. (2011). Pulsed out of awareness: EEG alpha oscillations represent a pulsed-inhibition of ongoing cortical processing. *Front Psychol*, 2, 10619. <https://doi.org/10.3389/FPSYG.2011.00099/BIBTEX>
- Miltner, W. H. R., Braun, C., Arnold, M., Witte, H., & Taub, E. (1999). Coherence of gamma-band EEG activity as a basis for associative learning. *Nature*, 397, 434–436. <https://doi.org/10.1038/17126>
- Minguillon, J., Lopez-Gordo, M. A., & Pelayo, F. (2016). Stress assessment by prefrontal relative gamma. *Front Comput Neurosci*, 10, Article 195054. <https://doi.org/10.3389/FNCOM.2016.00101/BIBTEX>
- Nater, U. M., La Marca, R., Florin, L., Moses, A., Langhans, W., Koller, M. M., & Ehlert, U. (2006). Stress-induced changes in human salivary alpha-amylase activity – associations with adrenergic activity. *Psychoneuroendocrinology*, 31, 49–58. <https://doi.org/10.1016/J.PSYNEUEN.2005.05.010>
- Nater, U. M., Rohleder, N., Gaab, J., Berger, S., Jud, A., Kirschbaum, C., & Ehlert, U. (2005). Human salivary alpha-amylase reactivity in a psychosocial stress paradigm. *Int J Psychophysiol*, 55, 333–342. <https://doi.org/10.1016/J.IJPSYCHO.2004.09.009>
- Obrist, P. A. (1981). Cardiovascular Psychophysiology. *Cardiovascular Psychophysiology*. <https://doi.org/10.1007/978-1-4684-8491-5>
- Perrin, S. L., Jay, S. M., Vincent, G. E., Sprajcer, M., Lack, L., Ferguson, S. A., & Vakulin, A. (2019). Waking qEEG to assess psychophysiological stress and alertness during simulated on-call conditions. *International Journal of Psychophysiology*, 141, 93–100. <https://doi.org/10.1016/J.IJPSYCHO.2019.04.001>
- Poppelaars, E. S., Klackl, J., Pletzer, B., & Jonas, E. (2021). Delta-beta cross-frequency coupling as an index of stress regulation during social-evaluative threat. *Biol Psychol*, 160, Article 108043. <https://doi.org/10.1016/J.BIOPSYCHO.2021.108043>
- Rönnerberg, J., Lunner, T., Zekveld, A., Sörqvist, P., Danielsson, H., Lyxell, B., Dahlström, Ö., Signoret, C., Stenfelt, S., Pichora-Fuller, M. K., & Rudner, M. (2013). The Ease of Language Understanding (ELU) model: Theory, data, and clinical implications. *Front Syst Neurosci*, 7, 48891. <https://doi.org/10.3389/FNSYS.2013.00031/BIBTEX>
- Schoofs, D., & Wolf, O. T. (2011). Are salivary gonadal steroid concentrations influenced by acute psychosocial stress? A study using the Trier Social Stress Test (TSST). *International Journal of Psychophysiology*, 80, 36–43. <https://doi.org/10.1016/J.IJPSYCHO.2011.01.008>
- Sica, C., & Ghisi, M. (2007). The Italian versions of the Beck Anxiety Inventory and the Beck Depression Inventory-II: Psychometric properties and discriminant power. *Leading-Edge Psychological Tests and Testing Research*, 27–50.
- Stemmler, G. (2004). Physiological Processes During Emotion. In P. Philippot, & R. S. Feldman (Eds.), *The Regulation of Emotion* (1st ed., pp. 38–69). New York: Psychology Press. <https://doi.org/10.4324/9781410610898>
- Strahler, J., Mueller, A., Rosenlocher, F., Kirschbaum, C., & Rohleder, N. (2010). Salivary alpha-amylase stress reactivity across different age groups. *Psychophysiology*, 47, 587–595. <https://doi.org/10.1111/J.1469-8986.2009.00957.X>
- Tabachnick, B. G., & Fidell, L. S. (2007). *Experimental designs using ANOVA*. CA: Thomson/Brooks/Cole Belmont.
- Thayer, J. F., Åhs, F., Fredrikson, M., Sollers, J. J., & Wager, T. D. (2012). A meta-analysis of heart rate variability and neuroimaging studies: Implications for heart rate variability as a marker of stress and health. *Neurosci Biobehav Rev*, 36, 747–756. <https://doi.org/10.1016/J.NEUBIOREV.2011.11.009>
- Vanhollebeke, G., De Smet, S., De Raedt, R., Baeken, C., van Mierlo, P., & Vanderhasselt, M. A. (2022). The neural correlates of psychosocial stress: A systematic review and meta-analysis of spectral analysis EEG studies. *Neurobiol Stress*, 18. <https://doi.org/10.1016/J.YNSTR.2022.100452>
- Wang, F., Wang, C., Yin, Q., Wang, K., Li, D., Mao, M., Zhu, C., & Huang, Y. (2015). Reappraisal writing relieves social anxiety and may be accompanied by changes in frontal alpha asymmetry. *Front Psychol*, 6. <https://doi.org/10.3389/FPSYG.2015.01604>
- Wróbel, A. (2000). Beta activity: A carrier for visual attention. *Acta Neurobiol Exp (Wars)*, 60, 247–260. <https://doi.org/10.55782/ANE-2000-1344>
- Yao, M., Lei, Y., Li, P., Ye, Q., Liu, Y., Li, X., & Peng, W. (2020). Shared Sensitivity to Physical Pain and Social Evaluation. *Journal of Pain*, 21, 677–688. <https://doi.org/10.1016/j.jpain.2019.10.007>